

CLOAK & DAGGER Master PCB Assembly Parts List

Designator	Description	Part No.
Capacitors		
C1-C9	0.1 μ F, +80% -20%, 50 V, Ceramic Capacitor	122002-104
C30	0.01 μ F, 100 V, Radial-Lead Mylar Capacitor	21-101103
C31	0.047 μ F, 100 V, Radial-Lead Mylar Capacitor	21-101473
C33	0.022 μ F, 100 V, Radial-Lead Mylar Capacitor	21-101223
C35	10 μ F, 25 V, Axial-Lead Aluminum Electrolytic Capacitor (Acceptable substitute is part no. 24-350106)	24-250106
C36-C38	0.1 μ F, $\pm$ 10%, 25 V, Ceramic Capacitor	122002-104
C39	0.01 μ F, 100 V, Radial-Lead Mylar Capacitor	21-101103
C40	0.022 μ F, 100 V, Radial-Lead Mylar Capacitor	21-101223
C42	0.022 μ F, 100 V, Radial-Lead Mylar Capacitor	21-101223
C44	1.0 μ F, 50 V, Axial-Lead Aluminum Electrolytic Capacitor	24-500105
C45	10 μ F, 25 V, Axial-Lead Aluminum Electrolytic Capacitor (Acceptable substitute is part no. 24-350106)	24-250106
C46	100 μ F, 25 V, Axial-Lead Aluminum Electrolytic Capacitor	24-250107
C47	0.1 μ F, $\pm$ 10%, 25 V, Ceramic Capacitor	122002-104
C51-C53	0.001 μ F, 100 V, Radial-Lead Mylar Capacitor	21-101102
C54, C55	10 μ F, 25 V, Axial-Lead Aluminum Electrolytic Capacitor (Acceptable substitute is part no. 24-350106)	24-250106
C56, C57	100 μ F, 35 V, Axial-Lead Aluminum Electrolytic Capacitor (Acceptable substitute is part no. 124001-107)	24-350107
C60	0.1 μ F, $\pm$ 10%, 25 V, Ceramic Capacitor	122002-104
C64-C66	100 pF, 100 V Minimum, Dipped Fixed Mica Capacitor	128002-101
C67-C109	0.1 μ F, $\pm$ 10%, 25 V, Ceramic Capacitor	122002-104
C111, C112	0.1 μ F, $\pm$ 10%, 25 V, Ceramic Capacitor	122002-104
C113	100 μ F, 25 V, Axial-Lead Aluminum Electrolytic Capacitor	24-250107
C114, C115	1.0 μ F, 50 V, Axial-Lead Aluminum Electrolytic Capacitor	24-500105
C116	0.1 μ F, $\pm$ 10%, 25 V, Ceramic Capacitor	122002-104
Coil		
L1-L4	1.0 μ H, 830 mA, 29 Ω , $\pm$ 10% Peaking Coil Inductor	141007-001
Diodes		
CR1-CR10	100 V Switching Diode	31-1N914
CR11	Red Light-Emitting Diode	38-MV5053
Integrated Circuits		
1A	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
1B	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
1C	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
1D/E	Type-74LS245 200 ns Tri-State Static RAM Integrated Circuit	137211-120
1F	Type-74LS245 Tri-State Octal Bus Transceiver Integrated Circuit	37-74LS245
1H	Type-74LS245 Tri-State Octal Bus Transceiver Integrated Circuit	37-74LS245
1J	Type-74LS139 Dual 2-4 Decoder/Multiplexer Integrated Circuit	37-74LS139
1L	Type-74LS27 Triple 3-Input NOR Gate Integrated Circuit	37-74LS27
1M	Type-74LS86 Quad 2-Input EXOR Gate Integrated Circuit	37-74LS86
1P	Type-74LS244 Line Driver/Receiver Integrated Circuit	37-74LS244
1R	Type-74LS244 Line Driver/Receiver Integrated Circuit	37-74LS244
2B	Type-74LS244 Line Driver/Receiver Integrated Circuit	37-74LS244

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CLOAK & DAGGER Master PCB Assembly Parts List, continued

Designator	Description	Part No.
2C	Type-74LS244 Line Driver/Receiver Integrated Circuit	37-74LS244
2D	EPROM Integrated Circuit	136023-301
2F	EPROM Integrated Circuit	136023-302
2H/J	EPROM Integrated Circuit	136023-303
2K	EPROM Integrated Circuit	136023-304
2L/M	Vertical Scroll Integrated Circuit	137321-1111
3C	Type-6502B 8-Bit Microprocessor Integrated Circuit	90-6012
3E/F	Type-74LS245 Tri-State Octal Bus Transceiver Integrated Circuit	37-74LS245
3F/H	Type-74LS245 Tri-State Octal Bus Transceiver Integrated Circuit	37-74LS245
3J/K	EPROM Integrated Circuit	137329-450
3L	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
3M	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
3N	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
3P	Type-74LS194 Shift Register Integrated Circuit	37-74LS194
3R	Type-74LS194 Shift Register Integrated Circuit	37-74LS194
4/5E	Type-74LS04 Hex Inverter Integrated Circuit	37-74LS04
4/5F	Type-74LS00 Quad 2-Input NAND Gate Integrated Circuit	37-74LS00
4/5H	Type-74LS245 Tri-State Octal Bus Transceiver Integrated Circuit	37-74LS245
4J	Type-74LS08 Quad 2-Input AND Gate Integrated Circuit	37-74LS08
4K	Type-74LS273 Octal D-Type Flip-Flop Integrated Circuit	37-74LS273
4L/M	200 ns Tri-State Static RAM Integrated Circuit	137211-120
4N	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
4P	Type-74LS194 Shift Register Integrated Circuit	37-74LS194
4R	Type-74LS194 Shift Register Integrated Circuit	37-74LS194
5B	Custom Integrated Circuit	C012294-01
5C/D	Custom Integrated Circuit	C012294-01
5E	Type-74LS04 Hex Inverter Integrated Circuit	37-74LS04
5E	Type-74LS138 Integrated Circuit	37-74LS138
5F	Type-74LS32 Quad 2-Input OR Gate Integrated Circuit	37-74LS32
5H	Type-74LS245 Tri-State Octal Bus Transceiver Integrated Circuit	37-74LS245
5J	Type-74LS20 Dual 4-Input NAND Gate Integrated Circuit	37-74LS20
5K	Type-74LS83 4-Bit Bin Full Adder Integrated Circuit	37-74LS83
5L	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
5M	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
5N	EPROM Integrated Circuit	136023-306
5R	EPROM Integrated Circuit	136023-305
6D	Type-74LS393 Dual 4-Bit Bin Counter Integrated Circuit	37-74LS393
6E	Type-74LS32 Quad 2-Input OR Gate Integrated Circuit	37-74LS32
6F	Type-74LS74 Dual D-Type Flip-Flop Integrated Circuit	37-74LS74
6H	Type-74LS174 Hex D-Type Flip-Flop Integrated Circuit	37-74LS174
6J	Type-74LS83 4-Bit Bin Full Adder Integrated Circuit	37-74LS83
6K	Type-74LS273 Octal D-Type Flip-Flop Integrated Circuit	37-74LS273
6L	250 ns Tri-State Static RAM Integrated Circuit	137250-001
6M	250 ns Tri-State Static RAM Integrated Circuit	137250-001
6N	Type-74LS32 Quad 2-Input OR Gate Integrated Circuit	37-74LS32
6P	Type-74LS194 Shift Register Integrated Circuit	37-74LS194
6R	Type-74LS194 Shift Register Integrated Circuit	37-74LS194
7A	Type-74LS14 Hex Inverter Schmitt-Trigger Integrated Circuit	37-74LS14

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CLOAK & DAGGER Master PCB Assembly Parts List, continued

Designator	Description	Part No.
7B	Type-74LS74 Dual D-Type Flip-Flop Integrated Circuit	37-74LS74
7C	Type-74LS00 Quad 2-Input NAND Gate Integrated Circuit	37-74LS00
7D	Type-74LS08 Quad 2-Input AND Gate Integrated Circuit	37-74LS08
7E	Type-74LS74 Dual D-Type Flip-Flop Integrated Circuit	37-74LS74
7F	Type-74LS74 Dual D-Type Flip-Flop Integrated Circuit	37-74LS74
7H	Type-74LS244 Line Driver/Receiver Integrated Circuit	37-74LS244
7J	Type-74LS163A 4-Bit Synchronous Counter Integrated Circuit	37-74LS163A
7K	Type-74LS163A 4-Bit Synchronous Counter Integrated Circuit	37-74LS163A
7L	Type-74LS163A 4-Bit Synchronous Counter Integrated Circuit	37-74LS163A
7M	Type-74LS163A 4-Bit Synchronous Counter Integrated Circuit	37-74LS163A
7N	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
7P	Type-74LS194 Shift Register Integrated Circuit	37-74LS194
7R	Type-74LS194 Shift Register Integrated Circuit	37-74LS194
8A	Type-74LS74 Dual D-Type Flip-Flop Integrated Circuit	37-74LS74
8B	Type-74LS32 Quad 2-Input OR Gate Integrated Circuit	37-74LS32
8C	Type-74S04 Hex Inverter Integrated Circuit	37-74S04
8D	Type-74LS32 Quad 2-Input OR Gate Integrated Circuit	37-74LS32
8E	Type-74LS00 Quad 2-Input NAND Gate Integrated Circuit	37-74LS00
8F	Type-74LS139 Dual 2-4 Decoder/Multiplexer Integrated Circuit	37-74LS139
8H	Type-74LS273 Octal D-Type Flip-Flop Integrated Circuit	37-74LS273
8J	Type-93422 45 ns Tri-State Bipolar Static RAM Integrated Circuit (Acceptable substitute is part no. 137178-001)	137178-002
8K	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
8L	Type-93422 45 ns Tri-State Bipolar Static RAM Integrated Circuit	137178-002
8M	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
8N	EPROM Integrated Circuit	136023-307
8R	EPROM Integrated Circuit	136023-308
9A	Type-74LS109A Integrated Circuit	37-74LS109A
9B	Type-74LS02 Quad 2-Input NOR Gate Integrated Circuit	37-74LS02
9C	Type-74LS20 Dual 4-Input NAND Gate Integrated Circuit	37-74LS20
9D	Type-74LS74 Dual D-Type Flip-Flop Integrated Circuit	37-74LS74
9E	Type-74LS139 Dual 2-4 Decoder/Multiplexer Integrated Circuit	37-74LS139
9F	Type-74LS08 Quad 2-Input AND Gate Integrated Circuit	37-74LS08
9H	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
9J	Type-74LS273 Octal D-Type Flip-Flop Integrated Circuit	37-74LS273
9K	Type-74S260 Dual 5-Input NOR Gate Integrated Circuit	37-74S260
9L/M	Open Collector Bipolar RAM Integrated Circuit	137322-001
9N	Type-74LS244 Line Driver/Receiver Integrated Circuit	37-74LS244
9R	Type-74LS244 Line Driver/Receiver Integrated Circuit	37-74LS244
10A	Type-7407 Hex Buffer/Driver Integrated Circuit	37-7407
10B	Type-74LS259 8-Bit Latch Integrated Circuit	37-74LS259
10C	Type-74LS00 Quad 2-Input NAND Gate Integrated Circuit	37-74LS00
10D	Type-74LS00 Quad 2-Input NAND Gate Integrated Circuit	37-74LS00
10E	Type-74LS04 Hex Inverter Integrated Circuit	37-74LS04
10F	Type-74LS86 Quad 2-Input Exclusive OR Gate Integrated Circuit	37-74LS86
10H	Type-74LS153 4 to 1 Data Selector/Multiplexer Integrated Circuit	37-74LS153
10J	Type-74LS153 4 to 1 Data Selector/Multiplexer Integrated Circuit	37-74LS153
10K	Type-74LS174 Hex D-Type Flip-Flop Integrated Circuit	37-74LS174

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CLOAK & DAGGER Master PCB Assembly Parts List, continued

Designator	Description	Part No.
10L	Type-74LS174 Hex D-Type Flip-Flop Integrated Circuit	37-74LS174
11C	Type-74LS74 Dual D-Type Flip-Flop Integrated Circuit	37-74LS74
11D	Type-74LS08 Quad 2-Input AND Gate Integrated Circuit	37-74LS08
11F	Type-74LS74 Dual D-Type Flip-Flop Integrated Circuit	37-74LS74
11H	Type-74LS139 Dual 2-4 Decoder/Multiplexer Integrated Circuit	37-74LS139
11J	Type-74LS157 Quad Data Selector/Multiplexer Integrated Circuit	37-74LS157
12B	Type-LM324 Quad Op-Amp Integrated Circuit	37-LM324
13B	512-Element Analog Delay Line Integrated Circuit	137310-001
13C/D	Type-LM324 Quad Op-Amp Integrated Circuit	37-LM324
13J	Type-TL082 Dual Operational Amplifier Integrated Circuit	37-TL082CP
Resistors		
R1	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
R2	68 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-683
R3	220 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-221
R4	22 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-223
R5	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R6-R8	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
R9-R16	220 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-221
R17-R24	330 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-331
R25-R28	220 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-221
R29-R32	330 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-331
R33, R34	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
R35-R41	470 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-471
R42-R48	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R65-R72	470 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-471
R73-R89	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R92-R94	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R95	2.2 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-222
R96	4.7 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-472
R97	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
R98, R99	100 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-101
R100-R102	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R103	2.2 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-222
R104	4.7 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-472
R105	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
R106, R107	100 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-101
R108-R110	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R111	2.2 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-222
R112	4.7 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-472
R113	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
R114, R115	100 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-101
R116	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R117	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
R118	18 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-180
R119	100 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-104
R120-R123	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R124-R127	470 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-471

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CLOAK & DAGGER Master PCB Assembly Parts List, continued

Designator	Description	Part No.
R128, R129	220 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-221
R130, R131	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R132–R139	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
R140	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R141, R142	1.8 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-182
R150–R152	22 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-223
R153	2.2 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-222
R154	68 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-683
R155	1.5 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-152
R156	12 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-123
R157	100 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-101
R158	180 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-184
R159	15 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-153
R162	15 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-153
R163–R166	22 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-223
R167	2.2 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-222
R169	150 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-151
R170–R176	1 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-102
R177	220 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-221
R178	330 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-331
R179	220 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-221
R180	330 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-331
R181	100 Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-101
R182, R183	6.8 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-682
R184	10 k Ω , $\pm 5\%$, $\frac{1}{4}$ W Resistor	110000-103
Sockets		
1D/E	24 Contact, Medium-Insertion-Force IC Socket	79-42C24
2D	28 Contact, Medium-Insertion-Force IC Socket	79-42C28
2F	28 Contact, Medium-Insertion-Force IC Socket	79-42C28
2H/J	28 Contact, Medium-Insertion-Force IC Socket	79-42C28
2K	28 Contact, Medium-Insertion-Force IC Socket	79-42C28
2L/M	24 Contact, Medium-Insertion-Force IC Socket	79-42C24
3C	40 Contact, Medium-Insertion-Force IC Socket	79-42C40
3J/K	24 Contact, Medium-Insertion-Force IC Socket	79-42C24
4L/M	24 Contact, Medium-Insertion-Force IC Socket	79-42C24
5B	40 Contact, Medium-Insertion-Force IC Socket	79-42C40
5C/D	40 Contact, Medium-Insertion-Force IC Socket	79-42C40
5N	24 Contact, Medium-Insertion-Force IC Socket	79-42C24
5R	24 Contact, Medium-Insertion-Force IC Socket	79-42C24
6L	22 Contact, Medium-Insertion-Force IC Socket	79-42C22
6M	22 Contact, Medium-Insertion-Force IC Socket	79-42C22
8J	22 Contact, Medium-Insertion-Force IC Socket	79-42C22
8L	22 Contact, Medium-Insertion-Force IC Socket	79-42C22
8N	24 Contact, Medium-Insertion-Force IC Socket	79-42C24
8R	24 Contact, Medium-Insertion-Force IC Socket	79-42C24
9L/M	28 Contact, Medium-Insertion-Force IC Socket	79-42C28

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CLOAK & DAGGER Master PCB Assembly Parts List, continued

Designator	Description	Part No.
Switch		
5A	8-Toggle DIP Switch	66-118PIT
Transistors		
Q1-Q3	350 mW, 60 V, Switching NPN Transistor	34-2N3904
Q4, Q5	8A, 80 V, NPN Darlington Transistor	34-2N6044
Miscellaneous		
	Test Points (Acceptable substitute is part no. 020670-01)	179051-002
	Interconnect Printed-Circuit Board—includes 60-pin connector, part no. 179153-160	A039431-01